

Abubakr Elmallah

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EDUCATION

University of California, Irvine

Irvine, CA

Bachelor of Science – Computer Science (3.90 GPA)

Expected June 2028

- **Awards:** Apple Swift Student Challenge Winner (1 of 350 worldwide), Congressional App Challenge Winner
- **Teaching:** Launched [STEM Heroes Academy](#)'s first App Development course as its **youngest instructor** at 17, teaching 30+ students (2024–2026)
- **Relevant Coursework:** Data Structures & Algorithms, Software Engineering, Data Management (SQL), Artificial Intelligence, Information Retrieval, Object-Oriented Programming, Discrete Mathematics

WORK EXPERIENCE

Lead Software Developer

February 2026 – Present

ICS Student Council (ICSSC) | studentcouncil.ics.uci.edu

Irvine, CA

- Lead a **12-person engineering team** building **PeterPlate**, ICSSC's open-source campus dining platform, **live in production** on web and iOS, serving menu, allergen, nutrition, and ratings data.
- Built a type-safe frontend with **Next.js (React)**, **TypeScript**, **Zustand**, **Tailwind CSS**, and **shadcn/ui**.
- Shipped a **scalable** serverless backend by pairing **tRPC** with **Drizzle ORM** on a 13-table **PostgreSQL** schema, deployed to **AWS Lambda** and **RDS** via **GitHub Actions** CI/CD with **Docker** + **Testcontainers** tests.

Lead Mobile & Backend Developer

March 2025 – Present

UCI Student Center & Event Services | studentcenter.uci.edu

Irvine, CA

- Own and ship three App Store apps: **100K+ installs**, **3.2M+ sessions** ([UCI Now](#), [ZOTFinder](#), [Esports](#)).
- Shipped **UCI Esports** solo: **15K-line SwiftUI** client plus a **Node.js** backend on **AWS Lambda/DynamoDB** with role-based staff auth.
- Lead a **4+ person Android team**, running hiring, technical interviews, and code review.
- Own the **AWS** infrastructure and **Python** backends behind the department's apps.
- Cut load times **40%** rebuilding two apps from **UIKit** to **SwiftUI**, extending both to **Mac** and **Vision Pro**.
- Cut crashes **85%** (10.4K to under 1.5K/month) refactoring a decade-old codebase with **20+ dependencies**.

PROGRAMMING PROJECTS

[Star Wars Discord Bots](#) | *Python, C++20, Linux* **500+ Servers**

[GitHub](#) | 2024 – 2025

- Grew [Sabacc Droid](#) (Python, discord.py, 5K+ lines) and [Aurebesh Droid](#) (C++20, D++, CMake) to **500+ Discord servers** with an event-driven, concurrent 8-player engine spanning 4 rule variants.
- Sustained **99.9% uptime** on **Linux** with a **GitHub Actions** pipeline that builds and cuts a release on every push.

[Datapad & Aurebesh Translator](#) | *SwiftUI, React, PWA + Android* **Top 10 Entertainment** [GitHub](#) | 2023

- Published my first app at 17 and grew it into free and premium tiers earning **\$1K+** from **25K+ installs** at **99.9% crash-free**, spanning iPhone, iPad, Mac, and Watch.
- Wrote a **1,400-line computer vision engine** decoding photographed glyphs with no trained model: adaptive thresholding, connected-component segmentation, and multi-signal template matching.
- Ported the suite to **React 19** and **TypeScript**, shipping an offline **PWA** and an **Android** build via **Capacitor** from one codebase.

[Al-Islam App Suite](#) | *SwiftUI, Kotlin* **Congressional & Swift Student Challenge Winner** [GitHub](#) | 2023

- Built **three open-source apps** ([Al-Islam](#), [Al-Quran](#), [Al-Adhan](#)) for iPhone, iPad, Mac, and Watch, serving **15K+ users** across **55K+ sessions** at **99.8% crash-free**.
- Cut load times **50%** with offline **JSON** parsing, **SwiftSoup** scraping, and on-device caching, and maintain all three from one shared codebase with a **3-way-merge** sync tool that ports changes between them.

LANGUAGES & TECHNOLOGIES

Languages: TypeScript, JavaScript, Python, SQL, Java, C++ (C++20), Swift, Kotlin, Bash

Frameworks: React, Next.js, Node.js, tRPC, REST APIs, Drizzle ORM, Tailwind CSS, SwiftUI, UIKit

Tools: AWS (Lambda, RDS, DynamoDB), Docker, PostgreSQL, Git, GitHub Actions (CI/CD), Linux, Testcontainers, Agile/Scrum